

Math Ref: Hugget & Tod

Chp 2: Review of tensor alg.

$$\epsilon_{abcd} \epsilon^{abcd} = -24$$

$$*F_{ab} = -\frac{1}{2} \epsilon_{abcd} F^{cd} \quad | \quad **F_{ab} = -F_{ab}$$

Chp 3: Lorentz Spinors at a point:

$$\text{Spin}(4) \cong \text{SL}(2, \mathbb{C}) \times \overline{\text{SL}(2, \mathbb{C})}, \quad \text{SO}(4, \mathbb{R}) \cong \text{Sp}(4)/\mathbb{Z}_2$$

$$\text{SL}(4, \mathbb{R}) \cong \text{SL}(2, \mathbb{R}) \times \text{SL}(2, \mathbb{R})$$

* A vector in $\mathbb{M}^4$ is V^m which is in the rep $(\frac{1}{2}, \frac{1}{2})$. Thus, we can expand V^m in terms of its $\text{SL}(2, \mathbb{R}) \times \overline{\text{SL}(2, \mathbb{R})}$ indices:

$$\begin{aligned} \chi_\alpha \psi_{\dot{\alpha}} &= \delta_\alpha^{\dot{\beta}} \delta_{\dot{\alpha}}^{\beta} \chi_\beta \psi_{\dot{\alpha}} = +2(\sigma^m)_{\alpha\dot{\beta}} (\sigma_m)^{\dot{\beta}\beta} \chi_\beta \psi_{\dot{\alpha}} \\ &= \frac{1}{2} \sigma^m_{\alpha\dot{\alpha}} (\bar{\psi} \sigma_m \chi) \end{aligned}$$

Now, we have an isomorphism between $T^*\mathbb{M}^4 \cong \text{SL}(2, \mathbb{C})/\mathbb{Z}_2$

$$[\underline{X} \in \text{SL}(2, \mathbb{C}) :$$

$$\det(\underline{X}) = X^m \eta_{mn} X^m$$

$$\underline{X} = \bar{\sigma}_n X^m \quad | \quad X^m = \frac{1}{2} \text{tr}(\sigma^m \underline{X})$$

From the expansion above:

$$\underline{X}_{\dot{\alpha}\alpha} = (\sigma^m \underline{X})_{\dot{\alpha}\alpha} = \sigma^m_{\alpha\dot{\alpha}} V_m$$

$$\therefore V_m = \frac{1}{2} (\bar{\psi} \sigma_m \chi)$$

Naturally, we can view σ_m as a map from $\sigma: T^*\mathbb{M}^4 \rightarrow \text{SL}(2, \mathbb{C})$

$$\bar{\sigma}(V) = \bar{\sigma}^m V_m \in \text{SL}(2, \mathbb{C})$$

* We conclude, we can express all vector quantities by a pair of spinor indices:

$$V^{\alpha\dot{\alpha}} := \frac{1}{\sqrt{2}} \sigma^{\alpha\dot{\alpha}}_\mu V^\mu = \frac{1}{\sqrt{2}} \begin{pmatrix} V^0 + V^3 & V^1 - iV^2 \\ V^1 + iV^2 & V^0 - V^3 \end{pmatrix}$$

$$\alpha: (\frac{1}{2}, 0) \quad | \quad \dot{\alpha}: (0, \frac{1}{2})$$

negative

Positive chirality.

Note:

$$M_{ab} V^a V^b = 2 \det(V^{\alpha\dot{\alpha}})$$

* V^a is null iff $\det(V^{\alpha\dot{\alpha}}) = 0$.

In this case $V^{\alpha\dot{\alpha}}_{null} = a^\alpha \bar{a}_{\dot{\alpha}}$

• This gives us an unconstrained way to represent null vectors in $\mathbb{M}^4$: Any pair of Weyl spinors of opposite chirality define a null vector

↳ convention

$$a_\alpha = a^\beta \epsilon_{\beta\alpha}, \quad a^\beta = \epsilon^{\beta\alpha} a_\alpha$$

∴

$$V_{\alpha\dot{\alpha}} = V^{\beta\dot{\beta}} \epsilon_{\beta\alpha} \epsilon_{\dot{\beta}\dot{\alpha}} = \frac{1}{\sqrt{2}} \begin{pmatrix} v^0 - v^3 & -(v^1 + i v^2) \\ -v^1 + i v^2 & v^0 + v^3 \end{pmatrix}$$

$$V^{\alpha\dot{\alpha}} V_{\alpha\dot{\alpha}} = 2 \det(V^{\alpha\dot{\alpha}}) = M_{ab} V^a V^b$$

$$\therefore M_{ab} \rightarrow \epsilon_{\alpha\dot{\alpha}} \epsilon_{\beta\dot{\beta}}$$

$$ds^2 = \epsilon_{\alpha\dot{\alpha}} \epsilon_{\beta\dot{\beta}} dx^{\alpha\dot{\alpha}} dx^{\beta\dot{\beta}}$$

$$M_{ab} \sigma^a_{\alpha\dot{\alpha}} \sigma^b_{\beta\dot{\beta}} = \epsilon_{\alpha\dot{\alpha}} \epsilon_{\beta\dot{\beta}}$$

• An important note on notation:

We define inner products on the spin bundles

S^+ , S^- w/ metric $\epsilon_{\alpha\beta}$ of $SL(2, \mathbb{C})$:

$$\langle K W \rangle := K^\alpha W_\alpha = K^\alpha W^\beta \epsilon_{\beta\alpha}$$

$$[\bar{K} \bar{W}] := \bar{K}^{\dot{\alpha}} \bar{W}_{\dot{\alpha}} = \bar{K}^{\dot{\alpha}} \bar{W}^{\dot{\beta}} \epsilon_{\dot{\beta}\dot{\alpha}}$$

* Take two null vectors in $\mathbb{M}^4$:



V^a_{null}, W^a_{null}

$$V^a_{null} = a^\alpha \bar{K}_{\dot{\alpha}}, \quad W^a_{null} = a^\alpha \bar{W}_{\dot{\alpha}}$$

$$\therefore V_{\alpha\dot{\alpha}} W^{\alpha\dot{\alpha}} = K^\alpha \bar{K}^{\dot{\beta}} W_\alpha \bar{W}_{\dot{\beta}} = -K^\alpha W_\alpha \bar{K}^{\dot{\alpha}} \bar{W}_{\dot{\alpha}} ?$$

→

*What are we doing? Well, we have complexified $M^4_{\mathbb{C}}$. The reason, as we shall see, is that we can obtain different flat spacetimes w/ different metric signatures. In much the same way that we obtain "Projective" theories in superspace, by imposing constraints of the Superfields, we shall impose different constraints on $\psi^{\alpha\dot{\alpha}}$ to obtain different theories based on

$$[\eta]_{ab} = \text{diag}(-1, 1, 1, 1)$$

$$\mathbb{H}: (0, 4) : (+ + + +)$$

$$M^4: (1, 3) \quad (- + + +)$$

$$K_4: (2, 2) \quad (- - + +) \text{ "Kleinian"}$$

Each has the rotation group $SO(p, q)$

Reality conditions:

(i) Lorentz Signature: Want to single out $M^4 \subset M^4_{\mathbb{C}} \cong M^4 \otimes \mathbb{C}$

Intuitively, we can simply obtain M^4 by requiring (x^0, x^1, x^2, x^3) to be real, hence

$$(x^{\alpha\dot{\alpha}})^{\dagger} := x^{\alpha\dot{\alpha}} \quad (\text{Hermitian})$$

$$(x^{\alpha\dot{\alpha}})^{\dagger} = \frac{1}{\sqrt{2}} \begin{pmatrix} \bar{x}^0 + \bar{x}^3 & \bar{x}^1 - i\bar{x}^2 \\ \bar{x}^1 + i\bar{x}^2 & \bar{x}^0 - \bar{x}^3 \end{pmatrix} \cong \begin{pmatrix} x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & x^0 - x^3 \end{pmatrix}$$

$$(x^{\alpha\dot{\alpha}})^{\dagger} = (\sigma^{\alpha\dot{\alpha}}_{\mu} V^{\mu})^{\dagger} = (\sigma^{\alpha\dot{\alpha}}_{\mu})^{\dagger} V^{\mu}$$

So, since $\exists$ a translation on $\sigma^{\alpha\dot{\alpha}}$ we expect

particular $(\sigma_m^{\alpha\dot{\alpha}})^T = \sigma_m^{\alpha\dot{\alpha}}$

- This reality condition is associated to complex conjugation of spinors in $\mathbb{M}^4$:

$$K^\alpha = (a, b) \in \mathbb{C}^2, \quad \bar{W}^{\dot{\alpha}} = (c, d) \in \mathbb{C}^2$$

$$K^\alpha \mapsto \bar{K}^{\dot{\alpha}} = (\bar{a}, \bar{b}), \quad \bar{W}^{\dot{\alpha}} \mapsto W^{\alpha} = (\bar{c}, \bar{d})$$

given a null vector $V_{null}^{\alpha\dot{\alpha}}$ we can write this as $V_{null}^{\alpha\dot{\alpha}} = K^\alpha \bar{K}^{\dot{\alpha}}$

$$\text{Since } \begin{pmatrix} a \\ b \end{pmatrix} \otimes (\bar{a} \ \bar{b}) = V_{null}^{\alpha\dot{\alpha}}$$

$$(V_{null}^{\alpha\dot{\alpha}})^\dagger = ((\bar{a} \ \bar{b})^\dagger \otimes \begin{pmatrix} a \\ b \end{pmatrix}^\dagger) = \begin{pmatrix} \bar{a} \\ \bar{b} \end{pmatrix} \otimes (a \ b) = V_{null}^{\alpha\dot{\alpha}} \quad \checkmark$$

Foldian Signature: For $x^{\alpha\dot{\alpha}}$ we define the operator:

$$\hat{X}^{\alpha\dot{\alpha}} := \frac{1}{\sqrt{2}} \begin{pmatrix} \bar{x}^0 - \bar{x}^3 & -\bar{x}^1 + i\bar{x}^2 \\ -\bar{x}^1 - i\bar{x}^2 & \bar{x}^0 + \bar{x}^3 \end{pmatrix} \quad \underline{\mathbb{R}^4}$$

$$\text{Demand } \hat{X}^{\alpha\dot{\alpha}} = x^{\alpha\dot{\alpha}}$$

$$\begin{pmatrix} \bar{x}^0 - \bar{x}^3 & -\bar{x}^1 + i\bar{x}^2 \\ -\bar{x}^1 - i\bar{x}^2 & \bar{x}^0 + \bar{x}^3 \end{pmatrix} = \begin{pmatrix} x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & x^0 - x^3 \end{pmatrix}$$

- $\bar{x}^0 = x^0 \Rightarrow x^0 \in \mathbb{R}$

- $\bar{x}^3 = -x^3, \quad \underline{x^3 = iy^3}$

- $-\bar{x}^1 + i\bar{x}^2 = x^1 - ix^2$

$$x^1 = -\bar{x}^1 \Rightarrow \underline{x^1 = iy^1} \quad \& \quad \underline{x^2 = iy^2}$$

$$(x^0, y^1, y^2, y^3) \in \mathbb{R}^4$$

$$\begin{pmatrix} x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & x^0 - x^3 \end{pmatrix} = \begin{pmatrix} x^0 + iy^3 & iy^1 + y^2 \\ iy^1 - y^2 & x^0 - iy^3 \end{pmatrix}$$

$$\therefore \boxed{X^{\alpha\dot{\alpha}}|_{x=\hat{x}} = \frac{1}{\sqrt{2}} \begin{pmatrix} x^0 + iy^3 & iy^1 + y^2 \\ iy^1 - y^2 & x^0 - iy^3 \end{pmatrix}}, \quad (x^0, y^1, y^2, y^3) \in \mathbb{R}^4$$

Note

$$2\det(x) = (x^0)^2 + (y^1)^2 + (y^1)^2 + (y^2)^2 = x^2$$

- This induces a conjugation on 2-spinors. This does not interchange spinor reps:

$$K^\alpha \mapsto \hat{K}^\alpha = (-\bar{b}, \bar{a}) \quad | \quad \bar{w}^{\dot{\alpha}} \mapsto \hat{\bar{w}}^{\dot{\alpha}} = (-\bar{d}, \bar{c})$$

This operation does not square to the identity

$$\hat{\hat{K}}^\alpha = (-a, -b) = -K^\alpha$$

$\therefore$ We need to apply $\sim$ of four-times to obtain K^α . Thus, the structure associated w/ $\mathbb{H}$ is Quaternionic.

- There is also non-trivial combination $K^\alpha \bar{w}^{\dot{\alpha}}$ which is preserved under $\sim$

$\Rightarrow$ There are no real null vectors in $\mathbb{H}$!

Split Signatures: $\mathbb{R}^{(2,2)}$

Here the operation is simply complex conjugation:

$$(x^{\alpha\dot{\alpha}})^* = x^{\alpha\dot{\alpha}}$$

$$x^{\alpha\dot{\alpha}}|_{\lambda=x^{\mu}} = \frac{1}{\sqrt{2}} \begin{pmatrix} x^0 + x^1 & x^1 + y^2 \\ x^1 - y^2 & x^0 - x^3 \end{pmatrix} \quad x^0, x^1, y^2, x^3 \in \mathbb{R}$$

$$2\det(x^{\alpha\dot{\alpha}}|_{\lambda=x^{\mu}}) = (x^0)^2 + (y^2)^2 - (x^1)^2 - (x^3)^2$$

* For Spinors, the complex conj. does not flip reps.

$$K^\alpha \mapsto (K^\alpha)^* = (a^*, b^*) \quad | \quad \bar{K}^{\dot{\alpha}} \mapsto (\bar{K}^{\dot{\alpha}})^* = (c^*, d^*)$$

- 2-Spinors in $\mathbb{R}^{2,2}$ are real valued:

$$\text{Spin}(4)^{\mathbb{C}} \subset \mathbb{R}^{2,2} \text{ is}$$

$$\text{Spin}(\mathbb{R}^{2,2})^{\mathbb{C}} \cong \text{SL}(2, \mathbb{R}) \times \text{SL}(2, \mathbb{R})$$

$$V_{\alpha\dot{\alpha}} = K^\alpha \bar{K}^{\dot{\alpha}}$$

$$K^\alpha, \bar{K}^{\dot{\alpha}} \in \mathbb{R}^2$$

Twistor space:

Let $\mathbb{CP}^3$: set of all complex lines through the origin of $\mathbb{C}^4$.

Homogeneous coords: $X^A = (x^1, \dots, x^4)$

s.t. $Z^A \neq 0$

This is simply a rep of $[X^A]$ where

$$y^A \in [X^A] \Rightarrow \exists \lambda \in \mathbb{C}^* \text{ s.t. } y^A = \lambda X^A$$

• Cover $\mathbb{CP}^3$ by charts

$$U_i = \{X^A \in \mathbb{C}^4 / x^i \neq 0\}$$

in U_i we can thus define

$$X_{(i)}^A = X^A / x^i$$

$$\text{on } U_{(1)}: (1, \frac{x^2}{x^1}, \frac{x^3}{x^1}, \frac{x^4}{x^1})$$

$$\text{For } x^1 = y, \quad x^i = y z^i$$

$$U_{(1)} = (1, z^2, z^3, z^4)$$

* Notes use Z^A , I'm using X^A with the decomp.
into $X^A = y Z^A$ so my notes use Z^A as

INHOMOGENEOUS: Z^A

HOMOGENEOUS: X^A

Def: We define Twistor space $\mathbb{PT}$ of $M^4_{\mathbb{C}}$
is an open subset of $\mathbb{CP}^3$

$$\mathbb{PT} \subset \mathbb{CP}^3$$

Split X^A into two Weyl spinors of opposite
chirality:

$$X^A = (\mu^{\dot{\alpha}}, \lambda_{\alpha})$$

$$(\mu^{\dot{\alpha}})^{\dagger} = \lambda_{\alpha} \quad \text{and} \quad (\lambda_{\alpha})^{\dagger} = \mu^{\dot{\alpha}}$$

line elements for twistor space are elements of $\mathbb{PT}$ and

this decomposition and an element of spacetime.
 This is known as the twistor correspondence. This
 is an algebraic relation between X^A and $x^{\alpha\dot{\alpha}}$
 $X^A \in \mathbb{PT}$, $x^{\alpha\dot{\alpha}} \in \mathbb{M}_{\mathbb{C}}^4$

These are the incidence relations

$$\mu^{\dot{\alpha}} = x^{\alpha\dot{\alpha}} \lambda_{\alpha}$$

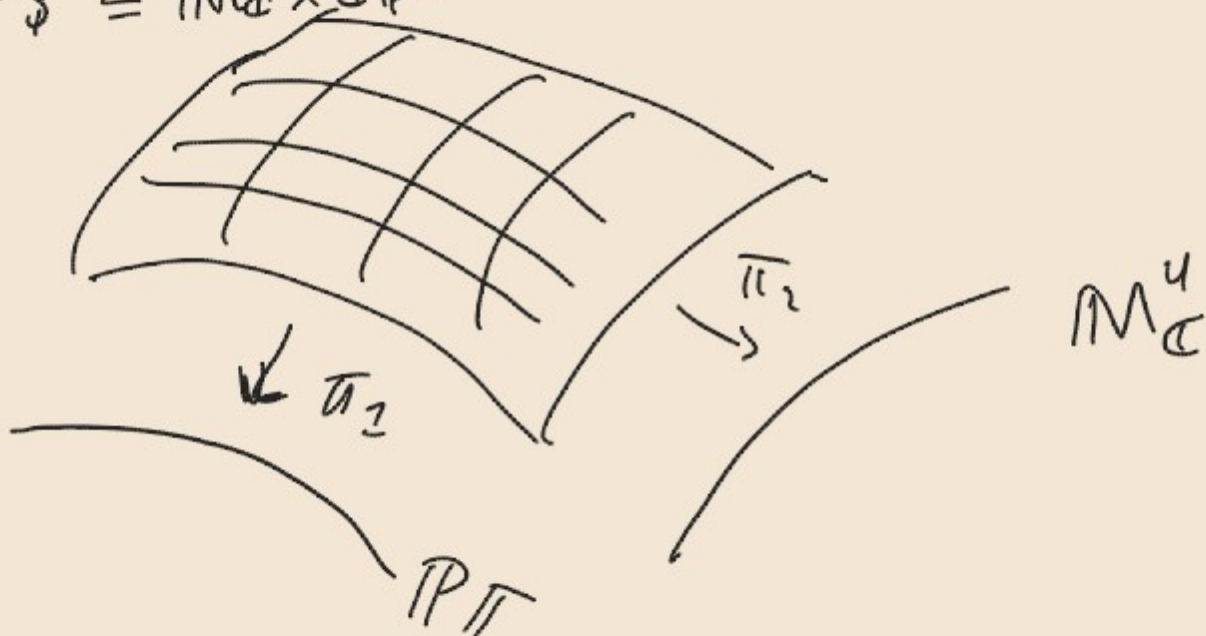
* As we will see in the Maths: This relationship
 is a result of a Double fibration of the

Projective spin bundle over $\mathbb{M}_{\mathbb{C}}^4$ and $\mathbb{PT}$

$(x^{\alpha\dot{\alpha}}, [\lambda_B])$ $\mathbb{PS}$ $(x^{\alpha\dot{\alpha}}, [\lambda_B])$

$\downarrow$ π_1 π_2 $\downarrow \pi_2$
 $(x^{\beta\dot{\alpha}}, \lambda_p, \lambda_{\alpha})$ $\mathbb{PT}$ $\mathbb{M}_{\mathbb{C}}^4$ $x^{\alpha\dot{\alpha}}$

$$\mathbb{PS} \cong \mathbb{M}_{\mathbb{C}}^4 \times \mathbb{CP}^1$$



* For most physical purposes we can we can simply
 think of $\mathbb{PT}$ simply by the relation
 $\mu^{\dot{\alpha}} = x^{\alpha\dot{\alpha}} \lambda_{\alpha}$

~> Quickly note that we can write this

using an Automorphism

$$f: \mathbb{P}\mathbb{S} \rightarrow \mathbb{P}\mathbb{S}$$

$$\mathbb{P}\mathbb{S} \xrightarrow{f} \mathbb{R}\mathbb{S}$$

$$\begin{array}{ccc} \pi_1 \downarrow & & \downarrow \pi_2 \\ \mathbb{P}\mathbb{T} & \longrightarrow & \mathbb{M}_0^4 \end{array}$$

What does $\mu^{\dot{\alpha}} = x^{\alpha\dot{\alpha}} \lambda_{\alpha}$ tell us?

(i) Suppose we fix a pt $x \in \mathbb{M}_0^4$, what is this in $\mathbb{P}\mathbb{T}$? $x^{\alpha\dot{\alpha}}$ are coeff's of a linear eq. relating $\mu^{\dot{\alpha}}$ & λ_{α} . Suppose, we forget about the projective scale of the coordinates. Then $X^A = (\mu^{\dot{\alpha}}, \lambda_{\alpha}) \in \mathbb{C}^4$

thus, the incidence relation $\mu^{\dot{\alpha}} = x^{\alpha\dot{\alpha}} \lambda_{\alpha}$ define a complex plane $\mathbb{C}^2 \subset \mathbb{C}^4$

* Recall, in $\mathbb{R}^3$, a plane is defined by $ax + by + cz + d = 0$

Adding the projective scale back in. $\mathbb{C}^2 \rightarrow \mathbb{CP}^2$
 $\mathbb{CP}^2 \subset \mathbb{P}\mathbb{T}$

Since the eq. is linear and holomorphic, we see a pt. in $\mathbb{M}_0^4$ corresponds to a linear & Holomorphically Embedded Riemann Sphere in Twistor space.



* Any holomorphic linear embedding of $\mathbb{CP}^2 \cong S^2 \setminus \{\infty\}$ into $\mathbb{CP}^3$ can always be put into the form of an incidence relation for $x^{\alpha\dot{\alpha}}$.

Q: Is this the result of Kodaira's theorem stating that given both conditions of proj. spaces they can be expressed algebraically?

If $\sigma_a := (\sigma_0, \sigma_1)$ are homogeneous coordinates of $\mathbb{CP}^1$

$$\mu^{\dot{\alpha}} = x^{\alpha\dot{\alpha}} \lambda_{\alpha}$$

We can expand λ_{α} in terms of these coords:

$$\lambda_{\alpha} = C_{\alpha}^a \sigma_a \quad \& \quad x^{\alpha\dot{\alpha}} C_{\alpha}^a \equiv b^{\dot{\alpha}a}$$

$$\mu^{\dot{\alpha}} = b^{\dot{\alpha}a} \sigma_a$$

Here $(b^{\dot{\alpha}a}, C_{\alpha}^a)$: 8-parameters which define a map between (coordinate changes)?

→ This is over counting, we must take the automorphism group of S^2

Naturally, S^2 has $SU(2)$ symmetry. We find there are 4 D.O.F:

③: from Automorph. of $\mathbb{CP}^1$: $SL(2, \mathbb{Z})$
the Möbius transform.

+ ① from $\lambda \in \mathbb{C}^*$
= 4

• Note: we have an $SL(2, \mathbb{R})$ symm since the sphere has lines which pierce the S^2 twice. Thus, we must use the Double cover $SL(2, \mathbb{R})/\mathbb{Z}_2$.
Not too sure how this becomes $SL(2, \mathbb{Z})$

Guess what find out as we go through the 'maths'.

We use these Gauge fixing to set $C_a^a = \delta_a^a$

$$\bar{m}^a = \bar{b}^{\dot{a}a} \sigma_a \quad \parallel \quad \lambda_a = \delta_a^{\dot{a}} \sigma_a$$

This is simply the $\mathbb{CP}^2$ vielbein formalism, gauge fixed.

- For $x \in M^4_{\mathbb{C}}$, the corresponding $S^2 \subset \mathbb{PT}$ by $L_x \cong \mathbb{CP}^1 \subset \mathbb{PT}$

We will refer to L_x as a "line"

$L_x \cong \mathbb{CP}^1$: "Line L_x associated to $x \in M^4_{\mathbb{C}}$ "

Since they are linearly embedded and defined holomorphically

Idea: a pt in $M^4_{\mathbb{C}}$ corresponds to an extended object in $\mathbb{PT}$ (twistor space)!

- We now want to ask: What happens when we have a pt in $\mathbb{PT}$, what is this in $M^4_{\mathbb{C}}$?

→ Let $X \in \mathbb{PT} \subset \mathbb{CP}^3$ be the intersection of two "lines." (holo, linear, embedded Riemann surfaces)

L_x and L_y

$$L_x \cap L_y = \{X \in \mathbb{PT}\}$$

$$\Rightarrow \mu^{\dot{a}} = x^{\alpha\dot{\alpha}} \lambda_{\alpha} \quad \text{and} \quad \mu^{\dot{a}} = y^{\alpha\dot{\alpha}} \lambda_{\alpha}$$

for $x, y \in M^4_{\mathbb{C}}$

Taking the difference:

$$0 = (x - y)^{\alpha\dot{\alpha}} \lambda_{\alpha}$$

Note, for non-trivial $X \neq Y$, we must have
 $(X-Y)^{\alpha\dot{\alpha}} \propto \lambda^{\alpha}$

Not too sure why, perhaps $\lambda^2 = 0$?

$$\therefore (X-Y)^{\alpha\dot{\alpha}} = \lambda^{\alpha} \tilde{\lambda}^{\dot{\alpha}}$$

$\therefore (X-Y)^{\alpha\dot{\alpha}}$ is a null vector! So X and Y are
null separated.

* Lines in $\mathbb{PT}$ (twistor space) intersect iff their
 corresponding pts. in $\mathbb{M}_0^4$ are null-separated.

• The pt $X \in \mathbb{PT}$ is described in $\mathbb{M}_0^4$ by varying
 over the choice of the spinor $\tilde{\lambda}^{\dot{\alpha}}$. Thus, this
 is a 2-plane which is totally null:

every tangent vector to the plane is of the
 form $\lambda^{\alpha} \tilde{\lambda}^{\dot{\alpha}}$, λ^{α} is fixed by the undotted
 components of $X \in \mathbb{PT}$

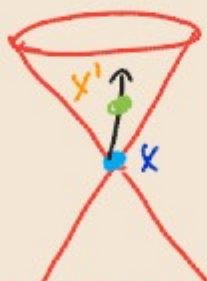
• These λ^{α} -planes are referred to as α -planes

We reiterate:

(i) A pt in $\mathbb{PT}$ is an α -plane in $\mathbb{M}_0^4$

Point in $\mathbb{M}_0^4$ corresponds to a linearly embedded Riemann
 sphere in twistor space

$\mathbb{M}_0^4$



$\mathbb{PT}$



- Points lying on the light cone $x \in IM_0$ are uniquely identified in $PT\mathbb{P}$ by lines which intersect $X \subset PT\mathbb{P}$

* Holomorphic structures in $PT\mathbb{P}$ encode conformal structures on spacetime.

- A holomorphic line in 3-complex dim is uniquely specified by any two pts which lie on that line.

Let X_1 and X_2 be two pts in $PT\mathbb{P}$ which lie on the line L_X



This implies? that we can represent the line L_X by the skew product

$$X_1 \wedge X_2 = \tilde{X}$$

$$X_2 = X_{(1)}^A dX_{(1)A}$$

Here $\tilde{X}$ is a "bi-twistor".

$$\tilde{X} = \frac{1}{2} X_1^A X_2^B d(X_1)_A \wedge d(X_2)_B$$

Recall, X^A is a coord. of PP^3 so in the expansion

$$X^A = (\mu^{\dot{\alpha}}, \lambda_{\alpha})$$

$$\tilde{X}^{AB} := X_1^A X_2^B$$

Not sure about this, but

$$\begin{aligned} \tilde{X}^{AB} &= \frac{1}{2} \left(\begin{pmatrix} \mu_1^{\dot{\alpha}} \\ \lambda_{\alpha}^{(1)} \end{pmatrix} \otimes \begin{pmatrix} \mu_2^{\dot{\beta}} \\ \lambda_{\beta}^{(2)} \end{pmatrix} - \begin{pmatrix} \mu_2^{\dot{\alpha}} \\ \lambda_{\alpha}^{(2)} \end{pmatrix} \otimes \begin{pmatrix} \mu_1^{\dot{\beta}} \\ \lambda_{\beta}^{(1)} \end{pmatrix} \right) \\ &= \frac{1}{2} \begin{pmatrix} \mu_1^{\dot{\alpha}} \mu_2^{\dot{\beta}} & \mu_1^{\dot{\alpha}} \lambda_{\beta}^{(2)} \\ \lambda_{\alpha}^{(1)} \mu_2^{\dot{\beta}} & \lambda_{\alpha}^{(1)} \lambda_{\beta}^{(2)} \end{pmatrix} - \begin{pmatrix} \mu_2^{\dot{\alpha}} \mu_1^{\dot{\beta}} & \mu_2^{\dot{\alpha}} \lambda_{\beta}^{(1)} \\ \lambda_{\alpha}^{(2)} \mu_1^{\dot{\beta}} & \lambda_{\alpha}^{(2)} \lambda_{\beta}^{(1)} \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} X_1^{\dot{\alpha}\dot{\beta}} X_2^{\beta\gamma} (\lambda_{\alpha}^{(1)} \lambda_{\beta}^{(2)} - \lambda_{\alpha}^{(2)} \lambda_{\beta}^{(1)}) & X_1^{\dot{\alpha}\dot{\beta}} (\lambda_{\alpha}^{(1)} \lambda_{\beta}^{(2)} - \lambda_{\alpha}^{(2)} \lambda_{\beta}^{(1)}) \\ X_2^{\dot{\alpha}\dot{\beta}} (\lambda_{\alpha}^{(1)} \lambda_{\beta}^{(2)} - \lambda_{\alpha}^{(2)} \lambda_{\beta}^{(1)}) & (\lambda_{\alpha}^{(1)} \lambda_{\beta}^{(2)} - \lambda_{\alpha}^{(2)} \lambda_{\beta}^{(1)}) \end{pmatrix} \end{aligned}$$

$$\begin{aligned}
 \therefore E_{EC} E_{AD} E^{CD} &= -E_{EC} \delta_A^C \delta_B^D \\
 \delta_E^D E_{AB} &= -E_{EC} \delta_A^C \delta_B^D = -E_{EC} (\delta_A^C \delta_B^D - \delta_A^D \delta_B^C) \\
 \delta_E^D E_{AD} &= - (E_{EA} \delta_D^D - E_{EB} \delta_A^D) \\
 E_{\alpha\beta} \langle \lambda^{(1)} | \lambda^{(2)} \rangle &= \frac{1}{2} (\lambda_{\alpha}^{(1)} \lambda_{\beta}^{(2)} - \lambda_{\beta}^{(1)} \lambda_{\alpha}^{(2)}) \\
 \text{proof: } \frac{1}{2} E_{\alpha\beta} \lambda_{\alpha}^{(1)} \lambda_{\beta}^{(2)} &= \frac{1}{2} E_{\alpha\beta} \epsilon^{\gamma\delta} \lambda_{\delta}^{(1)} \lambda_{\gamma}^{(2)} \\
 &= \left(-\delta_{\alpha}^{\gamma} \delta_{\beta}^{\delta} \lambda_{\delta}^{(1)} \lambda_{\gamma}^{(2)} \right) = -\frac{1}{2} (\delta_{\alpha}^{\gamma} \delta_{\beta}^{\delta} - \delta_{\beta}^{\gamma} \delta_{\alpha}^{\delta}) \lambda_{\delta}^{(1)} \lambda_{\gamma}^{(2)} \\
 &= -\frac{1}{2} (\lambda_{\beta}^{(1)} \lambda_{\alpha}^{(2)} - \lambda_{\alpha}^{(1)} \lambda_{\beta}^{(2)}) = \frac{1}{2} (\lambda_{\alpha}^{(1)} \lambda_{\beta}^{(2)} - \lambda_{\beta}^{(1)} \lambda_{\alpha}^{(2)}) \\
 \therefore \hat{X}^{AB} &= \langle \lambda^{(1)} | \lambda^{(2)} \rangle \begin{pmatrix} X^{\alpha\beta} E_{\alpha\beta} X^{\beta\delta} & X_{\beta}^{\alpha} \\ X_{\alpha}^{\beta} & E_{\alpha\beta} \end{pmatrix} \\
 \bullet \epsilon^{\alpha\gamma} \epsilon^{\delta\beta} X_{\gamma\delta} E_{\alpha\beta} X^{\beta\delta} &= \delta_{\gamma}^{\delta} \epsilon^{\alpha\delta} X_{\gamma\delta} X^{\beta\delta} = \epsilon^{\alpha\delta} X_{\gamma\delta} X^{\beta\delta} \\
 \bullet \epsilon^{\alpha\beta} X^2 &= X^{\alpha\gamma} X_{\gamma\delta} \epsilon^{\alpha\beta} \epsilon^{\gamma\delta} X^{\delta\delta} X^{\gamma\gamma} \\
 &= -2 \delta_{\gamma}^{\delta} \delta_{\delta}^{\beta} \epsilon^{\gamma\delta} X^{\delta\delta} X^{\gamma\gamma} \\
 &= -(\delta_{\gamma}^{\delta} \delta_{\delta}^{\beta} - \delta_{\gamma}^{\beta} \delta_{\delta}^{\delta}) \epsilon^{\gamma\delta} X^{\delta\delta} X^{\gamma\gamma} \\
 &= -(X^{\delta\beta} X^{\gamma\gamma} - X^{\gamma\beta} X^{\delta\delta}) \epsilon^{\gamma\delta}
 \end{aligned}$$

Twistor Geometry:

• We work with $TM_{\mathbb{C}}^4 \cong S \oplus \bar{S}$

$$\partial_m \leftrightarrow \partial_{\alpha\beta} := \frac{\partial}{\partial x^{\alpha\beta}}$$

On the spin bundles S and $\bar{S}$ we have symplectic forms $\epsilon_{\alpha\beta}$ & $\epsilon_{\dot{\alpha}\dot{\beta}}$

• A choice $TM_{\mathbb{C}}^4 \cong S \oplus \bar{S}$ amounts to choosing a conformal (holomorphic) structure.

$\triangleright$ A conf. structure is a line bundle L in

$$T^*M_{\mathbb{C}}^4 \oplus T^*M_{\mathbb{C}}^4$$

$$L \longrightarrow T^*M_{\mathbb{C}}^4 \oplus$$

• Consider the projectivisation of the dual bundle $(\bar{S})^*$

$$P(\bar{S}^*) \longrightarrow M_{\mathbb{C}}^4$$

is a $\mathbb{CP}^2$ -bundle over M^4 . Since S is rank two.
 $P(S^*)$ is (complex) 5-dimensional manifold which is
 bi-holomorphic to $\mathbb{CP}^4 \times \mathbb{CP}^1$

$$P(S^*) \cong F_5 \cong \mathbb{CP}^4 \times \mathbb{CP}^1$$

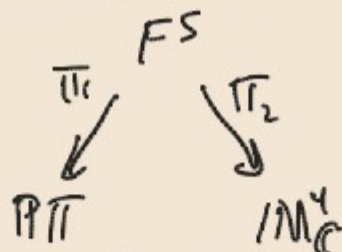
(Correspondence Space)

take this space to have coordinates
 $(x^{\alpha\dot{\alpha}}, \lambda_2)$

• The tangent space to F^S we have

$$V_\alpha = \lambda^\beta \partial_{\alpha\dot{\beta}} = \lambda^\beta \frac{\partial}{\partial x^{\alpha\dot{\beta}}}$$

→ Integrable rank-2-distribution on PF^S , i.e. a rank 2
 subbundle of TPF^S : a twistor distribution.



Since $P^2 \cong \mathbb{CP}^3$ so it can be covered in two
 charts $M^{\dot{\alpha}} = x^{\alpha\dot{\alpha}} \lambda_\alpha$

$$U_+ : (\lambda_1 \neq 0) \quad M_+^{\dot{\alpha}} := \frac{M^{\dot{\alpha}}}{\lambda_1}, \quad \lambda_+ := \frac{\lambda_2}{\lambda_1}$$

$$U_- : (\lambda_2 \neq 0) \quad M_-^{\dot{\alpha}} := M^{\dot{\alpha}} / \lambda_2, \quad \lambda_- := \lambda_1 / \lambda_2$$

$$\text{on } U_+ \cap U_- : M_+^{\dot{\alpha}} = \lambda_+ M_-^{\dot{\alpha}}$$

This covering means we can view this twistor space
 as $P^2 \cong \mathbb{CP}^3$ as the holomorphic fibration:

$$\mathcal{O}(2) \oplus \mathcal{O}(2) \rightarrow \mathbb{CP}^1 \quad \left(\begin{array}{l} \text{Resolved Conifold} \\ \text{Dual} \end{array} \right)$$

The tautological line bundle $\mathcal{O}(-1)$ (over $\mathbb{CP}^1$)
 is defined by

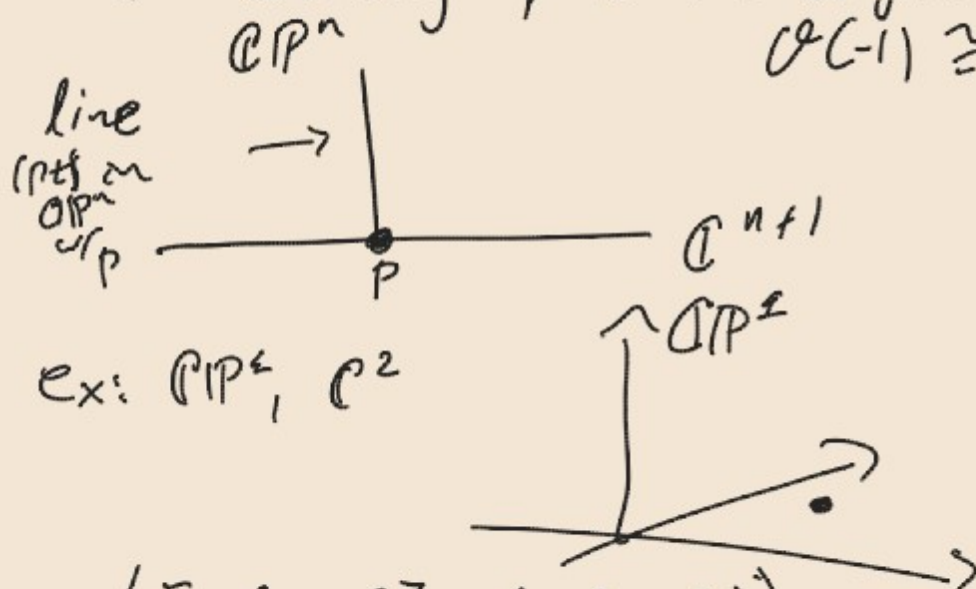
$$\mathcal{O}(-1) := \{ (\lambda_\alpha, \rho_\alpha) \in \mathbb{CP}^1 \times \mathbb{CP}^1 \mid \rho_\alpha \propto \lambda_\alpha \}$$

More concretely $\mathcal{O}(-1)$ can be the bundle

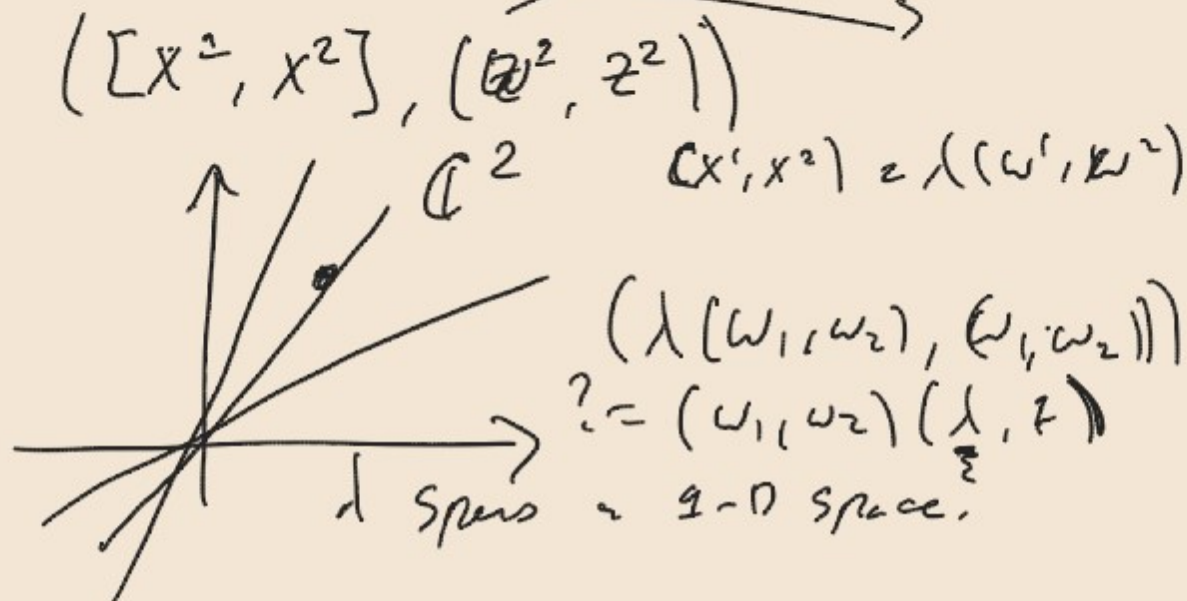
$([z_0, \dots, z_n], (\omega_0, \dots, \omega_n)) \in \mathcal{O}(-1)$, then $\exists \lambda \in \mathbb{C}^*$
 $(\omega_0, \dots, \omega_n) = \lambda [z_0, \dots, z_n]$

$\mathcal{O}(-1)$: this is the assignment of the line through the origin of $\mathbb{C}^{n+1}$ as defined in $\mathbb{CP}^n$. so each fiber of $\mathcal{O}(-1)$ over p is simply the line in $\mathbb{CP}^n$ connecting p to the origin.

$$\mathcal{O}(-1) \cong \mathbb{CP}^n \times \mathbb{CP}^{n+1}$$



ex: $\mathbb{CP}^1, \mathbb{C}^2$



$$([x^1, x^2], (\omega^1, \omega^2))$$

$$(x^1, x^2) = \lambda (\omega^1, \omega^2)$$

$$(\lambda (\omega_1, \omega_2), (\omega_1, \omega_2))$$

$$= (\omega_1, \omega_2) \left(\frac{\lambda}{\omega_1}, \frac{\lambda}{\omega_2} \right)$$

1 spans a 1-D space.

The Dual to this is the Hyperplane bundle $\mathcal{O}(1) = \mathcal{O}(-1)^*$

• The incidence relation $m^a = x^a \lambda_a$ identifies $x \in M^4$ w/ holomorphic sections of the Resolved Conifold.

$$\mathbb{P}^1 \cong \mathbb{CP}^3 / \mathbb{CP}^1$$

where the deletion is from $m^a \neq 0$ and $\lambda_a = 0$

- Klein correspondence: $\text{Pt } x \in M^4_q \rightarrow L_x := \pi_1(\pi_2^{-1}(x)) \hookrightarrow P\pi$

Twistor Geometry:

- After imposing the constraints on M^4_q of $\underline{x}$ to obtain different spacetimes we would like to know how this constraint is applied in $P\pi \cong \mathbb{CP}^3$

(1) Lorentz Signature: M^4

Recall that $\underline{x} = \underline{x}^* \Rightarrow \lambda^\alpha \mapsto \bar{\lambda}^{\dot{\alpha}}$
 $\bar{\mu}^{\dot{\alpha}} \mapsto \mu^\alpha$

So a twistor in PS

$$X^A = (\bar{\mu}^{\dot{\alpha}}, \lambda_\alpha) \mapsto (\bar{\lambda}_{\dot{\alpha}}, \bar{\mu}^\alpha)$$

- This sends a twistor in $PS \cong M^4_q \times \mathbb{CP}^1$ with elements in the complement rep.

This is projective Duality.

In Lorentz signature we modify the incidence relation as

$$\bar{\mu}^{\dot{\alpha}} = i x^{\alpha\dot{\alpha}} \lambda_\alpha$$

Let $P\pi^V$ be an open subset of $\mathbb{CP}^3$ w/
homogeneous coords

$$W_A = (\bar{\lambda}_{\dot{\alpha}}, \bar{\mu}^\alpha)$$

- Points in this Dual twistor space are related to points in M^4_c by

$$\bar{\mu}^\alpha = -i x^{\alpha\dot{\alpha}} \bar{\lambda}_{\dot{\alpha}}$$

$$\begin{array}{ccc} & P\mathcal{S}^* & \\ \downarrow & & \downarrow \\ P\pi & & M^4_q \end{array}$$

- Here we work w/ the dual $\mathcal{O}(1) \oplus \mathcal{O}(1)$
Note the total space $\mathcal{O}(1) \oplus \mathcal{O}(1) \rightarrow \mathbb{CP}^1$

is free dimensional $(\mathbb{C}(1) \oplus \mathbb{C}(1)) \times \mathbb{H}$.
 we have a natural inner product on
 $\mathbb{P}\mathbb{S} \times \mathbb{P}\mathbb{S} \rightarrow \mathbb{R} \leftarrow \mathbb{C}?$

or $\mathbb{P}\mathbb{S} \cong \mathbb{P}\mathbb{S}^V$

Let $X^A = (\mu^{\dot{\alpha}}, \lambda_{\alpha}) \in \mathbb{P}\mathbb{S}$

$W_A = (\bar{\lambda}_{\dot{\alpha}}, \bar{\mu}^{\alpha}) \in \mathbb{P}\mathbb{S}^V$

then

$$X \cdot W \equiv X^A W_A = [\mu \bar{\lambda}] + \langle \bar{\mu} \lambda \rangle$$

So inner products on $\mathbb{P}\mathbb{S}$ are induced by $SL(2, \mathbb{C})$
 invariant inner products on $\mathbb{T}\mathbb{S}$ and $\mathbb{T}\mathbb{S}^A$

* Complex conjugation of $\underline{x}$ in $IM_{\mathbb{C}}^4$ maps a
 twistor X^A to a pt in the dual space
 $\bar{X}_A$ whose components are given above. This
 induces? the inner product:

$$X \cdot \bar{X} = [\mu \bar{\lambda}] + \langle \bar{\mu} \lambda \rangle$$

When viewed as an inner product in $\mathbb{P}^n$ we find
 it has signature $(+2, -2)$

$$X \cdot \bar{X} = \mu^{\dot{\alpha}} \bar{\lambda}_{\dot{\alpha}} + \bar{\mu}^{\alpha} \lambda_{\alpha} = -i x^{\alpha \dot{\alpha}} \lambda_{\alpha} \bar{\lambda}_{\dot{\alpha}} + i (x^{\alpha \dot{\alpha}} \bar{\mu}_{\dot{\alpha}} \lambda_{\alpha})$$

* Since real Lorentzian spinors are valued in $SU(2)$, this
 means the inner product is an $SU(2, 2)$ -invariant.

• Given a line $L_X \cong \mathbb{CP}^1$ in $\mathbb{PT}$. {Could this line
 not just be $\mathcal{O}(-1)$?} The corresponding
 spacetime pt $x^{\alpha \dot{\alpha}}$ is valued in IM^4 ?

$$X \cdot \bar{X} = (x - x^{\dagger})^{\alpha \dot{\alpha}} \bar{\lambda}_{\dot{\alpha}} \lambda_{\alpha}$$

and $X \in M^4$ iff $X = X^\dagger$

- Any line L_X which corresponds to a pt. in M^4 must be contained in

$$PIN := \{X \in P\mathbb{T} \mid X \cdot \bar{X} = 0\}$$

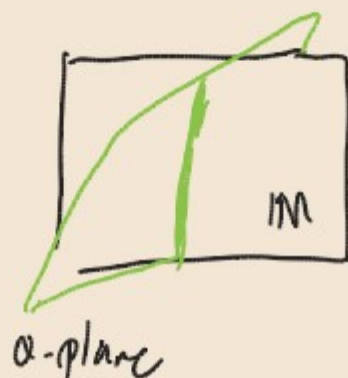
$PIN \subseteq P\mathbb{T}$ is the twistor space associated w/ M
"Space of null twistors"

* Recall that a pt. in $P\mathbb{T}$ is an α -plane in M^4 : totally null complex 2-plane whose tangent vectors are all proportional to λ^α .

Q: What is a pt. in PIN correspond to in M^4 ?

$X \cdot \bar{X} = 0$ singles out a single tangent vect. to the α -plane, namely $\lambda^\alpha \bar{\lambda}^\alpha$

$X \in PIN \longrightarrow$ Real null geodesic $\lambda^\alpha \bar{\lambda}^\alpha$



Euclidean Signature: We obtained a quaternionic

conjugation of spinors for $M^4 \rightarrow \mathbb{R}^4$

$$k^\alpha = (a, b) \mapsto \tilde{k}^\alpha = (-\bar{b}, \bar{a})$$

On a twistor $X^A = (\mu^{\dot{\alpha}}, \lambda_\alpha)$

$$\bar{X}^A = (x^{\alpha\dot{\alpha}} \hat{\lambda}_\alpha, \hat{\lambda}_\alpha)$$

This is an involution since $\sigma^2 = -I_{PT}$

• Naturally, no pt in $\mathbb{Z}^4$ is invariant under σ . A pt in PT is a totally null α -plane in IMQ and the statement that there are no real (wrt σ) pts in PT is the statement that this α -plane does not intersect the real slice $\mathbb{R}^4$ and there are no real null geodesics in positive-definite signature.

• A line in PT can be represented by the bi-tensor $X^{AB} = \hat{z}_1^A \hat{z}_2^B$. A line of the form $\hat{z}^A \hat{z}^B$ is preserved under this hat operation

$$\therefore X^{AB} = \hat{X}^{AB}$$

The Euclidean reality conditions thus imply: $\forall \hat{X} \in PT$ is uniquely associated w/ a pt $x \in \mathbb{R}^4$ by taking the line passing through x and $\hat{X}$: $X^{AB} = x^A \hat{x}^B$

This is a CP^1 fibration

$$PT \longrightarrow \mathbb{R}^4$$

$$\lambda^\alpha \lambda_\alpha = 0$$

not sure why.

Given locally by

$$x^{\alpha\dot{\alpha}} = (\hat{\mu}^{\dot{\alpha}} \lambda^\alpha - \mu^{\dot{\alpha}} \hat{\lambda}^\alpha) / \langle \lambda \hat{\lambda} \rangle$$

$$\bullet \hat{z}^A \wedge \hat{z}^B, \quad z^A = (\mu^{\dot{\alpha}}, \lambda_\alpha), \quad \hat{z}^B = (\hat{\mu}^{\dot{\alpha}}, \hat{\lambda}_\alpha)$$

$$\begin{pmatrix} \mu^{\dot{\alpha}} \\ \lambda_\alpha \end{pmatrix} \otimes \begin{pmatrix} \hat{\mu}^{\dot{\beta}} \\ \hat{\lambda}_\beta \end{pmatrix} = \begin{pmatrix} \mu^{\dot{\alpha}} \hat{\mu}^{\dot{\beta}} & \mu^{\dot{\alpha}} \hat{\lambda}_\beta \\ \lambda_\alpha \hat{\mu}^{\dot{\beta}} & \lambda_\alpha \hat{\lambda}_\beta \end{pmatrix}$$

$$= \begin{pmatrix} \hat{\mu}^{\dot{\beta}} \\ \hat{\lambda}_\beta \end{pmatrix} \otimes \begin{pmatrix} \mu^{\dot{\alpha}} \\ \lambda_\alpha \end{pmatrix} = \begin{pmatrix} \hat{\mu}^{\dot{\beta}} \mu^{\dot{\alpha}} & \hat{\mu}^{\dot{\beta}} \lambda_\alpha \\ \hat{\lambda}_\beta \mu^{\dot{\alpha}} & \hat{\lambda}_\beta \lambda_\alpha \end{pmatrix}$$

Looks like the above comes from the top right term

$$\frac{1}{2}(\mu^{\dot{\alpha}} \hat{\lambda}_{\dot{\beta}} - \tilde{\mu}^{\dot{\alpha}} \lambda_{\dot{\beta}})$$

$$(\hat{Z}^{\dot{\alpha}} \tilde{Z}^{\dot{\beta}})_{12} \epsilon^{\alpha\beta} = \frac{1}{2}(\mu^{\dot{\alpha}} \hat{\lambda}^{\dot{\alpha}} - \tilde{\mu}^{\dot{\alpha}} \lambda^{\dot{\alpha}})$$

However, the expansion is also given by

$$\epsilon^{\alpha\beta} (X^{\dot{\alpha}\dot{\beta}})_{12} = \langle \lambda \hat{\lambda} \rangle (X^{\dot{\alpha}\dot{\beta}})_{12} \epsilon^{\alpha\beta}$$

$$\therefore X^{\dot{\alpha}\dot{\beta}} = \frac{1}{2\langle \lambda \hat{\lambda} \rangle} (\mu^{\dot{\alpha}} \hat{\lambda}^{\dot{\beta}} - \tilde{\mu}^{\dot{\alpha}} \lambda^{\dot{\beta}})$$

This is close enough, I suppose.

* In $\mathbb{R}^4$ a pt $X \in \mathbb{PT}$ can be specified by
 a point in $\mathbb{R}^4$ (a line which is preserved by σ)
 and a pt in $S^2 \cong \mathbb{CP}^1$.
 $\mathbb{PS}^4 \cong \mathbb{R}^4 \times \mathbb{CP}^1$

Split signatures $\mathbb{R}^{2,2}$

$$X^A = (\mu^{\dot{\alpha}}, \lambda_{\dot{\alpha}}) \mapsto \bar{X}^{\bar{A}} = (\bar{\mu}^{\dot{\alpha}}, \lambda^{\dot{\alpha}})$$

* Points in $\mathbb{PT}$ which are preserved under σ are real valued points of $\mathbb{PT}$ i.e.

$$\mathbb{PT}^{\mathbb{R}} \subset \mathbb{CP}^3$$

Take a line L_X . Then for any $Z \in L_X$

$$Z = \bar{Z} \iff (X - \bar{X})^{\dot{\alpha}\dot{\beta}} \lambda_{\dot{\beta}} = 0$$

$$X^{\dot{\alpha}\dot{\beta}} = \bar{X}^{\dot{\alpha}\dot{\beta}}$$

* Euclidean signature preserves the complex-projective features of the general complexified signature twistor

Correspondence

(Complex Structure)

(Conformal Structure)

- Conformal structure of M^4 is determined by intersection of holomorphic lines in $\mathbb{P}T$.

* An almost complex structure $J \in \Gamma(\text{End}(M))$, $J^2 = -\text{id}$.

$$T M^{\mathbb{C}} = T_M^{(1,0)} \oplus T_M^{(0,1)}$$

$$V^a \frac{\partial}{\partial x^a} = V^a \frac{\partial}{\partial z^a} \oplus V^{\bar{a}} \frac{\partial}{\partial \bar{z}^{\bar{a}}}$$

$$\Omega^k(M)_{\mathbb{C}} = \bigoplus_{p+q=k} \Omega^{p,q}(M)$$

We can view $\Omega^{p,q}(M)$ as a bundle where the sections are (p,q) forms

$$\omega(p) = \omega_{a_1 \dots a_p} \bar{\omega}_{\bar{a}_1 \dots \bar{a}_q} dz^{a_1} \wedge \dots \wedge dz^{a_p} \wedge d\bar{z}^{\bar{a}_1} \wedge \dots \wedge d\bar{z}^{\bar{a}_q}$$

d on Ω^k has the expansion

$$d = \partial \oplus \bar{\partial}$$

$$\bar{\partial} := d\bar{z}^{\bar{a}} \wedge \frac{\partial}{\partial \bar{z}^{\bar{a}}}$$

- * An almost complex structure $\bar{\partial}$ is integrable i.e. a complex structure if $\bar{\partial}^2 = 0$. This is equivalent to $N_S = 0$ by the Newlander-Nirenberg theorem.

$\mathbb{C}P^3$

$\rightarrow$ On $\mathbb{P}T$: $\bar{\partial} := d\bar{X}^{\bar{a}} \frac{\partial}{\partial \bar{X}^{\bar{a}}}$

Euclidean Complex structure: $\mathbb{R}^4 \times \mathbb{C}P^2$

$$T_{\mathbb{P}T}^{0,1} = \text{span} \left\{ \bar{\partial}_0 = \langle \lambda \hat{\lambda} \rangle \lambda^a \frac{\partial}{\partial \lambda^a}, \bar{\partial}_{\dot{\alpha}} = \lambda^a \frac{\partial}{\partial x^{a\dot{\alpha}}} \right\}$$

$$\Omega_{\mathbb{P}T}^{0,1} = \text{span} \left\{ \bar{e}^0 = \frac{\langle \lambda d\lambda \rangle}{\langle \lambda \hat{\lambda} \rangle^2}, \bar{e}^{\dot{\alpha}} = \hat{\lambda}_a dx^{a\dot{\alpha}} / \langle \lambda \hat{\lambda} \rangle \right\}$$

Note that $\bar{\partial}$ is a sum over elements of T^{01} and Ω^{01} tensored together.

Further, in $\mathbb{E}$ Signature

$$X^{\alpha 2} = (\hat{\pi}^{\alpha} \lambda^{\alpha} - \pi^{\alpha} \hat{\lambda}^{\alpha}) / \langle \lambda \hat{\lambda} \rangle$$

$$\bar{\partial} = \bar{e}^0 \partial_0 + \bar{e}^2 \bar{\partial}_{\hat{\alpha}} = \frac{\langle \hat{\lambda} d\hat{\lambda} \rangle}{\langle \lambda \hat{\lambda} \rangle} \lambda^{\alpha} \frac{\partial}{\partial \hat{x}^{\alpha}} + \frac{\hat{\lambda}_{\alpha} d\hat{x}^{\alpha} \lambda_{\alpha}}{\langle \lambda \hat{\lambda} \rangle} \frac{\partial}{\partial x^{\alpha}}$$

$$= \frac{\langle \hat{\lambda} d\hat{\lambda} \rangle}{\langle \lambda \hat{\lambda} \rangle} \lambda^{\alpha} \frac{\partial}{\partial \hat{x}^{\alpha}} + \frac{\hat{\lambda}_{\alpha} d\hat{x}^{\alpha} \lambda^{\alpha}}{\langle \lambda \hat{\lambda} \rangle} \frac{\partial}{\partial x^{\alpha}}$$

$$\left(\hat{\lambda}^{\beta} (d\hat{\lambda})_{\beta} \lambda^{\alpha} = (\hat{\lambda})_{\beta} \hat{\lambda}^{\alpha} \lambda^{\beta} = (d\hat{\lambda})_{\beta} \frac{1}{2} \epsilon^{\alpha\beta} \langle \lambda \hat{\lambda} \rangle \right) \\ \Rightarrow -\frac{1}{2} (d\hat{\lambda})^{\alpha}$$

Conformal Structure:

Knowing about light cones is only enough to identify the conf. structure of M up to Boundary conditions.

→ Knowing about the light cones only classifies conformally flat spacetimes.

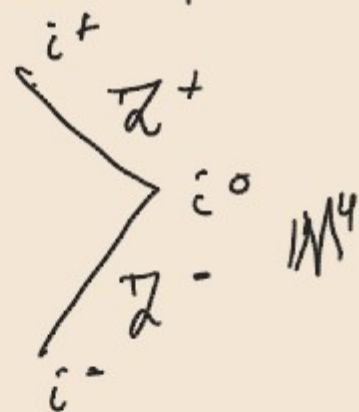
→ Must identify the difference between $M^4_{\mathbb{C}}$ and dS_4 by looking at "∞".

Conformal infinity $\mathcal{I}$

i^0 : Space-like infinity

$i^{\pm}$: Past/Future infinity

$\mathcal{I}^{\pm} \cong \mathbb{R} \times S^2$ Null Hypersurfaces



dS_4 : two space-like three spheres which form $i^{\pm}$

$$\text{Conf}(M^4_{\mathbb{C}}) \cong SL(4, \mathbb{C})$$

So M must have $SL(4, \mathbb{R})$ invariants. Recall, in 2D we obtain 2D invariants of $SL(2, \mathbb{R})$ via $E \wedge B$. Here: $E \wedge B \propto IP \wedge T \in \mathbb{CP}^3$

$$(z_1, z_2, z_3, z_4) := E_{ABCD} z_1^A z_2^B z_3^C z_4^D$$

• Let us now find linear reps of $SL(4, \mathbb{R})$:

$$\rho: SL(4, \mathbb{R}) \rightarrow \text{Aut}(SL(4, \mathbb{R}))$$

$$\rho(M) = M = \exp(\theta^A T_A)$$

$$z^A \mapsto \theta^B (T_B z)^A = \theta^B (T_B)^A_C z^C$$

$$T_B^A = z^A \frac{\partial}{\partial z^B}$$

*The generators of $SL(4, \mathbb{R})$ are given by
 $P_{\alpha\dot{\alpha}} = \lambda_\alpha \frac{\partial}{\partial x^{\dot{\alpha}}}$, $J_{\alpha\beta} = \lambda_\alpha \frac{\partial}{\partial \lambda^\beta}$, $\tilde{J}_{\dot{\alpha}\dot{\beta}} = \mu_{\dot{\alpha}} \frac{\partial}{\partial m^{\dot{\beta}}}$
 $K^{\alpha\dot{\alpha}} = \mu^{\dot{\alpha}} \frac{\partial}{\partial \lambda_\alpha}$, $D = \frac{1}{2} (\lambda_\alpha \frac{\partial}{\partial \lambda_\alpha} - \mu^{\dot{\alpha}} \frac{\partial}{\partial m^{\dot{\alpha}}})$

$$(i) P_m = \frac{\partial}{\partial x^m} \Rightarrow P_{\alpha\dot{\alpha}} = \frac{\partial}{\partial x^{\alpha\dot{\alpha}}}$$

In $\mathbb{E}^4$, ($\tilde{x}$ condition) we find that

$$P_{\alpha\dot{\alpha}} = \frac{\partial}{\partial x^{\alpha\dot{\alpha}}} = \frac{\partial \lambda_\beta}{\partial x^{\alpha\dot{\alpha}}} \frac{\partial}{\partial \lambda_\beta} + \frac{\partial \mu^{\dot{\beta}}}{\partial x^{\alpha\dot{\alpha}}} \frac{\partial}{\partial m^{\dot{\beta}}}$$

$$\Rightarrow \partial \lambda_\beta / \partial x^{\alpha\dot{\alpha}} = 0 \quad \text{similar to } x^I = y^J z^I$$

$$\text{and } \partial \mu^{\dot{\beta}} / \partial x^{\alpha\dot{\alpha}} = 0$$

$$\therefore \frac{\partial \mu^{\dot{\beta}}}{\partial x^{\alpha\dot{\alpha}}} = \frac{\partial x^{\beta\dot{\beta}}}{\partial x^{\alpha\dot{\alpha}}} \lambda_\beta = \delta_\alpha^\beta \delta_{\dot{\alpha}}^{\dot{\beta}} \lambda_\beta$$

$$P_{\alpha\dot{\alpha}} = \lambda_\alpha \frac{\partial}{\partial m^{\dot{\alpha}}}$$

$$\begin{aligned} (ii) D &= x^m \partial_m \rightarrow x^{\alpha\dot{\alpha}} \frac{\partial}{\partial x^{\alpha\dot{\alpha}}} \\ D &= \frac{1}{2} (\lambda_\alpha \frac{\partial}{\partial \lambda_\alpha} - \mu^{\dot{\alpha}} \frac{\partial}{\partial m^{\dot{\alpha}}}) \end{aligned}$$

- No way to distinguish between conformally flat spacetimes.

Any conformally flat metric can be written as

$$ds^2 = \Omega^2 dx^\alpha dx^\beta dx_\alpha dx_\beta$$

PT: (cannot distinguish between $f(x)=1$ and $f(x) \neq 1$)

pt in $M^4_0 \rightarrow$ Line L_x in PT.

x

$$x^{AB} = \langle \lambda_1, \lambda_2 \rangle \begin{pmatrix} \frac{1}{2} \epsilon^{\dot{\alpha}\dot{\beta}} x^2 x^{\dot{\alpha}}_1 x^{\dot{\beta}}_2 \\ -x^{\dot{\alpha}}_1 x^{\dot{\beta}}_2 \epsilon_{\dot{\alpha}\dot{\beta}} \end{pmatrix}$$

x^A_1 and x^B_2 are pts on the line $L_x \cong \mathbb{CP}^1$

$$ds^2 = \epsilon_{ABCD} dx^{AB} dx^{CD}$$

this is conformally flat

- If we want to interpret x^{AB} as spacetime coords, then we must consider them up to a scale. Thus, x^{AB} are homogeneous coords.
- Note this metric is simply the determinant of dx^{AB} :

$$\langle \rangle \begin{pmatrix} \epsilon^{\dot{\alpha}\dot{\beta}} x^{\dot{\alpha}}_i dx^{\dot{\beta}}_j & dx^{\dot{\alpha}}_j \\ -dx^{\dot{\alpha}}_i & 0 \end{pmatrix} \approx$$

$$\det(dx^{AB}) = \langle \rangle^2 dx^{\dot{\alpha}}_{\dot{\beta}} dx^{\dot{\beta}}_{\dot{\alpha}}$$

$$= \langle \rangle^2 \epsilon^{\dot{\alpha}\dot{\beta}} dx_{\dot{\beta}\dot{\gamma}} dx^{\dot{\gamma}\dot{\alpha}} \epsilon_{\dot{\alpha}\dot{\beta}} = \langle \rangle^2 \epsilon^{\dot{\alpha}\dot{\beta}} \epsilon_{\dot{\alpha}\dot{\beta}} dx_{\dot{\beta}\dot{\gamma}} dx^{\dot{\gamma}\dot{\alpha}}$$

ex: $\begin{pmatrix} A & B \\ C & D \end{pmatrix} \epsilon_{ABCD} \rightarrow \epsilon_{\dot{\alpha}\dot{\beta}\dot{\gamma}\dot{\delta}} \delta \dot{\alpha} \delta \dot{\beta} \delta \dot{\gamma} \delta \dot{\delta} \langle \rangle$

$$\rightarrow \epsilon_{\dot{\alpha}\dot{\beta}} \epsilon^{\dot{\beta}\dot{\alpha}} dx^{\dot{\alpha}}_{\dot{\gamma}} \epsilon_{\dot{\gamma}\dot{\beta}} \epsilon^{\dot{\beta}\dot{\gamma}} dx_{\dot{\alpha}\dot{\gamma}} \\ = \delta_{\dot{\alpha}\dot{\gamma}} \delta^{\dot{\gamma}\dot{\alpha}} dx^{\dot{\alpha}}_{\dot{\gamma}} dx_{\dot{\alpha}\dot{\gamma}} = \langle \rangle^2 dx^{\dot{\alpha}\dot{\beta}} dx_{\dot{\alpha}\dot{\beta}}$$

So, somehow, there should be extra factors of $(\epsilon_{\alpha\beta} \epsilon^{\beta\alpha})$.

$$ds^2 = \langle \lambda \lambda \rangle dx^{\alpha\beta} dx_{\alpha\beta}$$

Must rewrite $ds^2 = \epsilon_{ABCD} dx^{AB} dx^{CD}$ to be

$$ds^2 = (\epsilon_{ABCD} dx^{AB} dx^{CD}) / (I_{AB} X^{AB})^2$$

I_{AB} : fixed bispinor.

• Singular at $I_{AB} X^{AB} = 0$ which define points at infinity: Infinity twistor. Needed to be an open subset of $\mathbb{CP}^3$, rather than all of $\mathbb{CP}^3$

If $L_x \subset \mathbb{CP}^3$ and $I_{AB} X^{AB} = 0$ corresponds to points which lie at ∞ in spacetime, then they should not be included $\mathbb{PT}$.

Choose: $I_{AB} = \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 0 & \epsilon^{\alpha\beta} \end{pmatrix}$

$$\begin{aligned} \therefore I_{AB} X^{AB} &\cong \frac{1}{2} (000\epsilon^{\alpha\beta}) \begin{pmatrix} \frac{1}{2}\epsilon^{\alpha\beta} x^2 \\ x^{\dot{\alpha}\dot{\beta}} \\ -x^{\dot{\alpha}}_{\dot{\beta}} \\ \epsilon^{\alpha\beta} \end{pmatrix} \langle \lambda \lambda \rangle \\ &= \frac{\langle \lambda \lambda \rangle}{2} \epsilon^{\alpha\beta} \epsilon_{\alpha\beta} = \langle \lambda \lambda \rangle \end{aligned}$$

$$\therefore ds^2 = \frac{\epsilon_{ABCD} dx^{AB} dx^{CD}}{(\langle \lambda \lambda \rangle^2)}$$

$I_{AB} X^{AB} = \langle \lambda_1 \lambda_2 \rangle = 0$ implies that $\lambda_{2\alpha} \propto \lambda_{1\alpha}$

But $X^{AB} = X_1^A \wedge X_2^B$ where X_1^A and X_2^B both live in the same line. L_x in $\mathbb{PT}$. Here we must have $\lambda_{1\alpha} = \lambda_{2\alpha} = 0$

M_4

PT

Therefore the points of L^∞ must be of the form $X^A = (\mu^i, 0)$ however $\mu^i = x^{\alpha i} \lambda_\alpha$

So $\mu^i = 0$, since X^A are homogeneous

So X^A cannot equal zero: ($\mu^i = 0$ and $\lambda_\alpha = 0$)

$\Rightarrow$ Some component of $x^{\alpha i}$ must be ∞ at the pts where $\lambda_\alpha = 0$.

$\therefore$ Lines in PT for which $I_{AB} X^{AB} = 0$ should correspond to pts at ∞ in M_4 .

$$\epsilon^{ABCD} I_{AB} I_{CD} = 0$$

$\therefore$ The ∞ -twistor I_{AB} corresponds to a line L^∞ in PT . This line in PT is i^0 in M^4

Lines which intersect L^∞ corresponds to points in spacetime which are null separated from i^0 : $\mathcal{I}^I$

Spinors: Anti spinor: $\bar{\alpha}_\beta$

Sym: α_β

We also have $TM_4^C \cong S \oplus \bar{S}$, so we have symmetric exchanges of α and $\bar{\alpha}$. Not sure how this applies to $\Theta_\alpha \lambda^\alpha$

$$\phi_{\alpha\beta\gamma} = \phi_{\beta\alpha\gamma} = \phi_{\beta\gamma\alpha}$$

All spinors are null wrt. the $SL(2, \mathbb{C})$ metric $\epsilon_{AB} \lambda^A \lambda^B = 0$.

$$\text{If } \epsilon_{AB} \lambda^A \sigma^B = 0 \Rightarrow \sigma^B \propto \lambda^A$$

* The 2-dim case of a twistor is a line

Recall $e_a^\alpha \frac{\partial}{\partial x^\alpha} = e_\alpha$
 $g_{\mu\nu} = e_\mu^a g_{ab} e_\nu^b$
 $\therefore g_{ab} = (\sigma)_{\alpha i} (\sigma)_{\beta j} \xi^{\alpha\beta} \xi^{ij}$

The dyad: (σ_A, ι_A) is used to normalize
 $e_a^\mu g_{\mu\nu} e_b^\nu = \delta_{ab}$

Here we want:

$$E_{AB} \sigma^A \iota^B = \sigma_B \iota^B := 1$$

- Spinorial objects which are normalized. ι^B is the spinor and σ_B its dual in S^*

→ A dyad defines a (null) tetrad in M^4

$$l^a = \sigma^\alpha \bar{\sigma}^{\dot{\alpha}} ; n^a = \iota^\alpha \bar{\iota}^{\dot{\alpha}}$$

$$m^a = \sigma^\alpha \iota^{\dot{\alpha}} ; \bar{m}^a = \iota^\alpha \bar{\sigma}^{\dot{\alpha}}$$

l^a, n^a : real future pointing null vectors
 m^a : complex null vector.

Identifying g_{ab} w/ $E_{\alpha\beta} E_{\dot{\alpha}\dot{\beta}}$

$$g_{ab} l^a n^b = E_{\alpha\beta} E_{\dot{\alpha}\dot{\beta}} \sigma^\alpha \bar{\sigma}^{\dot{\alpha}} \iota^\beta \bar{\iota}^{\dot{\beta}}$$

$$= (E_{\alpha\beta} \sigma^\alpha \iota^\beta) (E_{\dot{\alpha}\dot{\beta}} \bar{\sigma}^{\dot{\alpha}} \bar{\iota}^{\dot{\beta}}) = 1$$

$$= E_{\alpha\beta} E_{\dot{\alpha}\dot{\beta}} (-\sigma^\alpha \iota^{\dot{\beta}}) (\iota^\beta \bar{\sigma}^{\dot{\alpha}})$$

$$= -g_{ab} m^a \bar{m}^b$$

- The tetrad is $\mathbb{Z}_2$ invariant under the ident. of a dyad $(\sigma^\alpha, \iota^\alpha)$

* In $SL(2, \mathbb{C})$ or 2-D $E_{AB} E_{CD} = 0$

$$E_{AB} E_{CD} - E_{AC} E_{BD} + E_{AD} E_{BC} - E_{AB} E_{DC} = 0$$

$$\rightarrow E_{AB} E_{CD} - E_{AC} E_{BD} - E_{AD} E_{CB}$$

$$\epsilon_{AB} \epsilon^{CD} = \epsilon_{AC} \epsilon_{BD} + \epsilon_{AD} \epsilon_{CB}$$

$$\epsilon_{AD} \delta_D^E = +\delta_A^E \epsilon_{BD} - \epsilon_{AD} \delta_B^E$$

$$\epsilon^{DG} \epsilon_{AB} \delta_D^E = \delta_A^E \epsilon^{DG} \epsilon_{BD} - \epsilon^{DG} \epsilon_{AD} \delta_B^E$$

$$\epsilon^{EG} \epsilon_{AB} = -\delta_A^E \delta_B^G + \delta_B^E \delta_A^G$$

$$\therefore \epsilon_{AD} \epsilon^{CD} = +\delta_A^C \delta_B^D - \delta_A^D \delta_B^C$$



(different to susy conventions)

As in supersymmetry, this is useful in that we can use it as so: Let $\phi \dots [CD] \dots$

$$\text{then } \epsilon_{AB} \epsilon^{CD} \phi \dots [CD] \dots$$

$$= +2 \delta_{AB}^{CD} \phi \dots [CD] \dots = 2 \phi \dots [AB]$$

$$\therefore \frac{1}{2} \epsilon_{AB} \phi \dots C \dots = \phi \dots [AB]$$

→ We can replace anti-sym indices by the contraction of a spinor of those indices times $\frac{1}{2} \epsilon$. Thus, we are effectively "removing" anti-sym. components leaving a symmetric spin-tensor.

• Symmetric spinors factorise: $\phi(\alpha, \alpha') = \alpha(\alpha_1 \dots \alpha_n)$

proof: Choose a dyad $\zeta^A = (1, X)$ and form

$$\phi_{A \dots B} \zeta^A \zeta^B = \phi_{1 \dots B} \zeta^B + \phi_{2 \dots B} X \zeta^B$$

work w/ ϕ_{AB} for simplicity:

$$\text{LHS} = \phi_{1B} \zeta^B + \phi_{2B} X \zeta^B = \phi_{11} + \phi_{12} X + \phi_{21} X + \phi_{22} X^2$$

$$\phi_{AB} \zeta^A \zeta^B = \phi_{11} + 2X \phi_{12} + \phi_{22} X^2$$

So we have a polynomial in X , here, Quadratic.

Since S is complex this factorises

$$\left[\frac{-2\phi_{12} \pm \sqrt{4\phi_{12}^2 - 4\phi_{11}\phi_{22}}}{2\phi_{22}} = -\phi_{12} \pm \sqrt{\phi_{12}^2 - \phi_{11}\phi_{22}} \right]$$

$$1 \dots \dots \dots$$

Take a general 2-tensor T_{ab}

$$\begin{aligned} T_{ab} &\mapsto T_{\alpha\dot{\alpha}\beta\dot{\beta}} = T_{a\dot{b}}\alpha_{\dot{\alpha}}\beta_{\dot{\beta}} = T(\alpha\dot{\beta})\alpha_{\dot{\alpha}} + T[\alpha\dot{\beta}]\alpha_{\dot{\alpha}}\beta_{\dot{\beta}} \\ &= T(\alpha\dot{\beta})\alpha_{\dot{\alpha}} + \frac{1}{2}\epsilon_{\alpha\beta}T_{\gamma\delta}\alpha_{\dot{\alpha}}\beta_{\dot{\beta}} \\ &= T(\alpha\dot{\beta})\alpha_{\dot{\alpha}} + \frac{1}{2}T(\alpha\dot{\beta})\gamma^{\delta}\epsilon_{\dot{\alpha}\dot{\beta}} + \frac{1}{2}(\epsilon_{\alpha\beta})(T_{\gamma\delta}(\alpha\dot{\beta}) + \frac{1}{2}\epsilon_{\dot{\alpha}\dot{\beta}}T) \\ &= T(\alpha\dot{\beta})(\alpha_{\dot{\alpha}}\beta_{\dot{\beta}}) + \frac{1}{2}\epsilon_{\dot{\alpha}\dot{\beta}}T(\alpha\dot{\beta})\gamma^{\delta} + \frac{1}{2}\epsilon_{\alpha\beta}T_{\gamma\delta}(\alpha\dot{\beta}) \\ &\quad + \frac{1}{4}\epsilon_{\alpha\beta}\epsilon_{\dot{\alpha}\dot{\beta}}T_{\gamma\delta}\gamma^{\gamma}\gamma^{\delta} \end{aligned}$$

(i) $T(\alpha\dot{\beta})(\alpha_{\dot{\alpha}}\beta_{\dot{\beta}})$: Hermitian spinor, corresponds to the symmetric trace-free part of T_{ab}

(ii) $\frac{1}{4}\epsilon_{\alpha\beta}\epsilon_{\dot{\alpha}\dot{\beta}}T_{\gamma\delta}\gamma^{\gamma}\gamma^{\delta}$: trace of T_{ab} since it is proportional to $g_{ab}\frac{1}{4}T^c_c$

(iii) $\left[\frac{1}{2}\epsilon_{\dot{\alpha}\dot{\beta}}T(\alpha\dot{\beta})\gamma^{\delta} + \frac{1}{2}\epsilon_{\alpha\beta}T_{\gamma\delta}(\alpha\dot{\beta})\right]$: Hermitian spinor which is $T[\alpha\dot{\beta}]$.

* Note, this implies that the two form:

$$F_{ab} \mapsto \partial_{\alpha\dot{\alpha}}A_{\beta\dot{\beta}} - \partial_{\beta\dot{\beta}}A_{\alpha\dot{\alpha}}$$

Choose anti-sym in $\alpha\dot{\alpha}\beta\dot{\beta}$, sym in $\alpha\dot{\beta}$ and sym under $S \leftrightarrow \bar{S}$. i.e. $[\alpha, \dot{\alpha}] = 0$

The trace of F_{ab} is zero since

$$g^{ab}F_{ab} = 0$$

and since $(\alpha\dot{\alpha}) = 0$ the first and last terms vanish.

$$\begin{aligned} F_{\alpha\dot{\alpha}\beta\dot{\beta}} &= \frac{1}{2}\epsilon_{\alpha\beta}T_{\gamma\delta}(\alpha\dot{\beta}) + \frac{1}{2}\epsilon_{\dot{\alpha}\dot{\beta}}T(\alpha\dot{\beta})\gamma^{\gamma} \\ &= \epsilon_{\alpha\beta}\tilde{F}_{\dot{\alpha}\dot{\beta}} + \epsilon_{\dot{\alpha}\dot{\beta}}F_{\alpha\beta} \end{aligned}$$

Here we define the Self-dual (SD) and anti-self dual (ASD) portions of the field strength.

$$\tilde{F}_{\alpha\beta} = \hat{F}(\alpha\beta) := \frac{1}{2} F_{\alpha\beta} \gamma_{\beta}$$

$$F_{\alpha\beta} = F(\alpha\beta) = \frac{1}{2} R_{\alpha\beta} \gamma_{\beta}$$

$$*E_{abcd} = i(E_{\alpha\gamma} E_{\beta\delta} E_{\gamma\delta} G_{\beta\gamma} - E_{\alpha\delta} E_{\beta\gamma} G_{\gamma\delta} E_{\beta\gamma})$$

$$E_{ab}{}^{cd} = i(\delta_{\alpha}^{\gamma} \delta_{\beta}^{\delta} \delta_{\gamma}^{\delta} \delta_{\delta}^{\gamma} - \delta_{\alpha}^{\delta} \delta_{\beta}^{\gamma} \delta_{\gamma}^{\delta} \delta_{\delta}^{\gamma})$$

$$\therefore F_{\alpha\beta\gamma\delta} = E_{\alpha\beta} \tilde{F}_{\gamma\delta} + E_{\gamma\delta} F_{\alpha\beta}$$

$$(*F)_{\alpha\beta} = \frac{1}{2} E_{\alpha\beta}{}^{cd} F_{cd}$$

$$= \frac{1}{2} i (\delta_{\alpha}^{\gamma} \delta_{\beta}^{\delta} \delta_{\gamma}^{\delta} \delta_{\delta}^{\gamma} - \delta_{\alpha}^{\delta} \delta_{\beta}^{\gamma} \delta_{\gamma}^{\delta} \delta_{\delta}^{\gamma}) (E_{\gamma\delta} \tilde{F}_{\alpha\beta} + E_{\alpha\beta} F_{\gamma\delta})$$

$$= \frac{1}{2} i (E_{\alpha\beta} F_{\gamma\delta} + F_{\alpha\beta} E_{\gamma\delta} \dots)$$

$$= -i F_{\alpha\beta} E_{\gamma\delta} + i \tilde{F}_{\alpha\beta} E_{\gamma\delta}$$

$\therefore$ We see that $\tilde{F}_{\alpha\beta}$ has eigenvalue $+i$ under

$*$ and $F_{\alpha\beta}$ has eigenvalue $-i$

$E_{\alpha\beta} F_{\gamma\delta}$: Self dual (SD)

$E_{\gamma\delta} F_{\alpha\beta}$: Anti-Self Dual (ASD)

$F = F^+ + F^-$, where

$$F^+ := \frac{1}{2} (F + i * F) \quad | \quad F^- = \frac{1}{2} (F - i * F)$$

Maxwell eq's $dF^+ = 0$

$$dF^- = 0$$

Bianchi: $\nabla_{[a} F_{bc]} = 0$

$$\nabla^a F_{ab} = 0 \Leftrightarrow \nabla^{a\dot{a}} E_{\alpha\beta} \tilde{F}_{\dot{\alpha}\dot{\beta}} = 0$$

$$E_{\alpha\beta} \nabla^{a\dot{a}} \tilde{F}_{\dot{\alpha}\dot{\beta}} = 0$$

$$\nabla^{a\dot{a}} (E_{\alpha\beta} \tilde{F}_{\dot{\alpha}\dot{\beta}} + E_{\dot{\alpha}\dot{\beta}} F_{\alpha\beta}) = 0$$

$$17 \nabla^a F_{ab} = \nabla^a F_{ab} = 0 \quad (Maxwell eq's)$$

$$\nabla_{[a} F_{bc]} = 0$$

$$\nabla_{[\alpha} (\epsilon_{\beta\gamma} \tilde{F}_{\delta\epsilon]} + \epsilon_{\delta\epsilon} F_{\beta\gamma}) = 0$$

$$\nabla_{\beta}^{\alpha} \tilde{F}_{\alpha\beta} - \nabla_{\beta}^{\alpha} F_{\alpha\beta} = 0 \quad (\text{Bianchi})$$

Bianchi is non-dynamical: Any F $\nabla_{[a} F_{bc]} = 0$. Together we see

$$\nabla F^+ = \nabla F^- = 0 \quad \text{SD} \oplus \text{ASD sol}^n$$

SD gauge field is characterised by

$$\tilde{F}_{\alpha\beta} = 0$$

With this constraint we are left with

$$\partial_{\beta}^{\alpha} \tilde{F}_{\alpha\beta} = 0 \quad (\text{Purely self dual relation})$$

Similarly for a purely ASD solⁿ. These

(SD $\oplus$ ASD): Solⁿs for 4D

These each characterise the two polarizations of the photon in 4D i.e. it characterises the two types of Helicity of the photon w/ Gauge field A_{μ} .

$$\begin{array}{ll} \text{SD:} & + \text{ Helicity } (F_{\alpha\beta} = 0) \\ \text{ASD:} & - \text{ Helicity } (\tilde{F}_{\alpha\beta} = 0) \end{array} \quad \begin{array}{l} \partial_{\beta}^{\alpha} \tilde{F}_{\alpha\beta} = 0 \\ \partial_{\beta}^{\alpha} F_{\alpha\beta} = 0 \end{array}$$

These are spin-2 zero mass equations

- The (Gauge-invariant) curvature tensor associated to the spin-s gauge field containing SD & ASD components

We can generalise this result by looking at the

Curvature of the 4D - spacetime:

$$R_{abcd} = G_{ab} \epsilon_{cd} \tilde{\Psi}_{\dot{\alpha}\dot{\beta}\dot{\gamma}\dot{\delta}} + \dots$$

$$= (\tilde{\Psi}, \psi, \phi, R)$$

$\tilde{\Psi}, \psi$: SD and ASD solⁿ of the Weyl curvature.

ϕ : trace free Ricci

R : Ricci scalar $R = \phi^{\dot{\gamma}\delta}_{\alpha\beta}$

• In a vacuum solⁿ

$$R_{\mu\nu} = \frac{1}{2} g_{\mu\nu} R$$

$$R = \frac{D}{2} R \Rightarrow \underline{R = 0}$$

• On the support of these equations the Bianchi identity $\nabla_{\mu} R_{\nu\rho\sigma\tau} = 0$ is equivalent to

$$\epsilon_{\dot{\gamma}\delta} \nabla_{\dot{\beta}} \tilde{\Psi}_{\dot{\alpha}\dot{\beta}\dot{\gamma}\dot{\delta}} - \epsilon_{\dot{\gamma}\delta} \nabla_{\dot{\beta}}^{\alpha} \psi_{\alpha\dot{\beta}\dot{\gamma}\dot{\delta}} = 0$$

so we obtain SD and ASD solⁿ as above with

$$\boxed{\nabla_{\dot{\beta}} \tilde{\Psi}_{\dot{\alpha}\dot{\beta}\dot{\gamma}\dot{\delta}} = 0} \quad | \quad \boxed{\nabla_{\dot{\beta}}^{\alpha} \psi_{\alpha\dot{\beta}\dot{\gamma}\dot{\delta}} = 0}$$

• These equations are linearized by taking

$$\nabla_{\dot{\beta}} \mapsto \partial_{\dot{\beta}}$$

The above is therefore given by helicity ± 2 particles: graviton.

* A general helicity h is represented by a field w/ $2|h|$ dotted or undotted symm spinor indices w/ E.O.M:

$$h > 0 : \tilde{\Phi}_{\dot{\alpha}_1 \dots \dot{\alpha}_{2|h|}} , \quad \partial^{\beta\dot{\alpha}_1} \tilde{\Phi}_{\dot{\alpha}_1 \dots \dot{\alpha}_{2|h|}} = 0$$

$$h = 0 : \phi , \quad \square \phi = \partial^{\alpha\dot{\alpha}} \partial_{\alpha\dot{\alpha}} \phi = 0$$

$Z_m \equiv (\text{zero-rest-mass})$

- This rep associates two totally symm. spinors with any field w/ spin $s \geq 0$. These spinors encode the information contained in the totally trace free portion of the linearised spin s tensor associated w/ the field.

Spin 1: F_{ab}

Spin 2: R_{abcd}

$$\Omega^2(M^4) = \Omega_+^2(M^4) \oplus \Omega_-^2(M^4)$$

under $\star$ eigenvalues

